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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 17

TITLE: Write a Java program to demonstrate the use of Swing and Applets

Problem Statement:

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

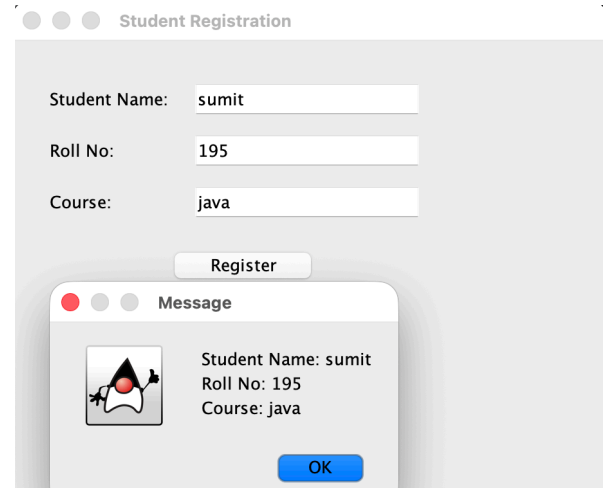
Theory

Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.

Exercise

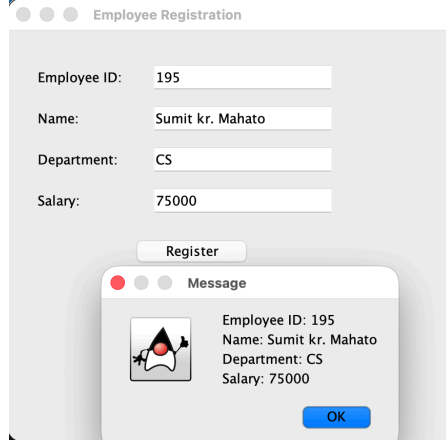
1. Create a simple Student Registration Form GUI using Swing.

```
→ abc
J StudentRegistration.java 1 X J Untitled-1 X J import javax.s...
sumit > Documents > java > java17 > J StudentRegistration.java > StudentRegistratic
1  import javax.swing.*;
2
3  public class StudentRegistration {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          JFrame frame = new JFrame("Student Registration");
9
10         JLabel label1 = new JLabel("Student Name:");
11         JLabel label2 = new JLabel("Roll No:");
12         JLabel label3 = new JLabel("Course:");
13
14         JTextField text1 = new JTextField();
15         JTextField text2 = new JTextField();
16         JTextField text3 = new JTextField();
17
18         JButton button = new JButton("Register");
19
20         label1.setBounds(30, 30, 100, 30);
21         text1.setBounds(140, 30, 180, 30);
22
23         label2.setBounds(30, 70, 100, 30);
24         text2.setBounds(140, 70, 180, 30);
25
26         label3.setBounds(30, 110, 100, 30);
27         text3.setBounds(140, 110, 180, 30);
28
29         button.setBounds(120, 160, 120, 30);
30
31         frame.add(label1);
32         frame.add(text1);
33         frame.add(label2);
34         frame.add(text2);
35         frame.add(label3);
36         frame.add(text3);
37         frame.add(button);
38
39         button.addActionListener(e -> {
40             JOptionPane.showMessageDialog(frame,
41                 "Student Name: " + text1.getText() +
42                 "\nRoll No: " + text2.getText() +
43                 "\nCourse: " + text3.getText());
44         });
45
46         frame.setLayout(null);
47         frame.setSize(370, 250);
48         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
49         frame.setVisible(true);
50     }
```



2.Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box.

```
Users > sumit > Documents > java > java17 > J EmployeeRegistration.java > Emplc
1  import javax.swing.*;
2
3  public class EmployeeRegistration {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          JFrame frame = new JFrame("Employee Registration");
9
10         JLabel label1 = new JLabel("Employee ID:");
11         JLabel label2 = new JLabel("Name:");
12         JLabel label3 = new JLabel("Department:");
13         JLabel label4 = new JLabel("Salary:");
14
15         JTextField text1 = new JTextField();
16         JTextField text2 = new JTextField();
17         JTextField text3 = new JTextField();
18         JTextField text4 = new JTextField();
19
20         JButton button = new JButton("Register");
21
22         label1.setBounds(30, 30, 100, 30);
23         text1.setBounds(140, 30, 180, 30);
24
25         label2.setBounds(30, 70, 100, 30);
26         text2.setBounds(140, 70, 180, 30);
27
28         label3.setBounds(30, 110, 100, 30);
29         text3.setBounds(140, 110, 180, 30);
30
31         label4.setBounds(30, 150, 100, 30);
32         text4.setBounds(140, 150, 180, 30);
33
34         button.setBounds(120, 200, 120, 30);
35
36         frame.add(label1);
37         frame.add(text1);
38         frame.add(label2);
39         frame.add(text2);
40         frame.add(label3);
41         frame.add(text3);
42         frame.add(label4);
43         frame.add(text4);
44         frame.add(button);
45
46         button.addActionListener(e -> {
47             JOptionPane.showMessageDialog(frame,
48                 "Employee ID: " + text1.getText() +
49                 "\nName: " + text2.getText() +
50                 "\nDepartment: " + text3.getText() +
51                 "\nSalary: " + text4.getText()
52             );
53
54             frame.setLayout(null);
55             frame.setSize(370, 300);
56             frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
57             frame.setVisible(true);
58         });
59     }
```



Conclusion:

The program successfully demonstrated the use of **Java Swing and Applets** for developing graphical user interfaces. Swing components such as `JFrame`, `JLabel`, `JTextField`, and `JButton` were used to create simple registration forms and handle user input. The exercise helped in understanding basic GUI design, event handling, and displaying information through dialog boxes.